

Paper #26b: Sterile Neutrino Mass Generation — UQFF (Redirect Notice)

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Session: Phase 1 (Sessions 1-43)

Framework: UQFF Star-Magic ($\kappa = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.61$)

Source: CondensedPhysics2.py, MAIN_1_CoAnQi.cpp

Cross-links: PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md (canonical), PAPER_027 (Lepton Flavor Violation), PAPER_028 (BSM Coupling Constants)

Abstract

This file is a redirect stub. All sterile neutrino mass generation content — including the GUT-scale sector, Yukawa coupling matrix, seesaw mechanism, neutrinoless double beta decay constraints, and UQFF vacuum-mediated mass generation equations — has been merged into the canonical whitepaper PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md. See that file for full derivations and numerical results.

Key parameters consolidated: sterile neutrino mass $M_N \sim 1e14$ GeV, active-sterile mixing $|U_{eN}|^2 < 1e-3$, vacuum contribution $\delta m_\nu = \kappa \times [\text{SSq}] \times m_N$.

MERGED INTO: PAPER_026_Sterile_Neutrino_Mass_Generation_UQFF.md

Key Equations (Summary)

The UQFF-modified seesaw mass formula:

$$m_\nu = m_D^2/M_N + \delta m_{UQFF}$$

$$\delta m_{UQFF} = \kappa \times [\text{SSq}] \times \frac{v^2}{M_N} = 5.0 \times 10^{-4} \times 0.57 \times \frac{v^2}{M_N}$$

$$|U_{eN}|^2 < 1.0 \times 10^{-3} \quad (\text{experimental constraint})$$

Numerical summary: $M_N = 1.0e14$ GeV, $m_D = 1.74e2$ GeV (top-quark scale), $\delta m_{UQFF} \approx 2.85e-4 \times v^2/M_N$, $\kappa[\text{SSq}] = 2.85e-4$

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Merged by: GitHub Copilot

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments. LHC direct-search results and arXiv constraints bound the parameter space.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete – 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{SCm} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.